

Laksh Goyal

La Jolla, CA, USA | +1 (858) 298 8940 | lakshgoyal06@gmail.com | linkedin.com/in/lakshgoyal

EDUCATION

University of California San Diego

B.S in Computer Science | GPA 3.7

San Diego, CA

Graduation June 2027

Relevant Coursework: Machine Learning; Artificial Intelligence; Software Engineering; Algorithms & Data Structures; Systems Programming; Computer Architecture; Linear Algebra; Probability & Statistics

EXPERIENCE

Engineers for Exploration

San Diego, CA

Machine Learning Engineer - Mangrove Monitoring ML

Oct 2025 – Present

- Improved boundary accuracy by 9% by adding a boundary-aware IoU metric for mangrove canopy segmentation
- Enabled configurable metric modes (segmentation vs. loss-only) for more controlled experiments and evaluation
- Refactored evaluation into a metric registry, decoupling metrics from the training loop to enable boundary-aware IoU
- Exposed errors masked by standard IoU and verified end-to-end workflows while maintaining backward compatibility

Software Engineer - Acoustic Data pipelines

Oct 2025 – Present

- Processed multi-terabyte acoustic data using ingestion + preprocessing pipelines for CNN and EfficientNet classifiers
- Increased usable audio segments by 15% by optimizing pipeline throughput and automating preprocessing
- Built indexing and query utilities for NAS-stored acoustic data to support faster training and evaluation cycles

Healthlink@UCSD

San Diego, CA

Technical Lead & Software Developer

Jun 2025 – Present

- Improved website load time by 20% by migrating Webflow components to a React-based frontend
- Developing membership tracker using React and Supabase to log events and update status for 100+ members

Computer Science & Engineering Society

San Diego, CA

Software Developer

Jun 2025 – Mar 2026

- Improved skin-lesion classification by 7% by fine-tuning MedGemma-1.5-4B-IT on a 25K+ labeled image dataset
- Cleaned and standardized medical image data by filtering low-quality samples and augmenting to improve robustness
- Ran inference and evaluation on 200+ validation images to validate real-time feedback in a cross-platform app

Xyntopia @ Taskyon

San Diego, CA

AI/ML Intern

Jun 2025 – Oct 2025

- Built a state-based LLM task framework in Python, routing tasks through planning, execution and completion
- Reduced task retries by 35% using validation and state-aware partial retries instead of rerunning full tasks
- Collaborated in a shared Git-based codebase to integrate LLM tools with JSON/YAML interfaces, improving usability

PROJECTS

Helmet Identification Model | *Azure ML, Blob Storage, OpenCV, Python*

- Built a multi-stage computer vision pipeline detecting motorbike riders and classifying helmet usage using ResNet-50
- Trained on 5K+ street images, achieving 94% accuracy and 0.92 mAP to support traffic law enforcement
- Optimized for real-time deployment, achieving < 50 ms GPU inference while balancing accuracy-latency tradeoffs
- Published a research paper through the Lumiere Research Program, ranked Top 10% of 200 submissions

PipeScoutX | *Python, Flask, React, PostgreSQL, Docker*

- Built a pipe-inspection robot integrating sensors with C/Arduino/ESP32 and using YOLOv3 for crack detection
- Used OpenCV contour analysis for localization & optimized inference by tuning thresholds on embedded hardware
- Achieved 90% crack-detection accuracy while balancing accuracy-latency tradeoffs under real-time constraints
- Validated across 100+ meters of pipeline, landing 2nd at World Robotic Olympiad India for reliability & performance

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, Assembly, Flutter

Frameworks/Libraries: React, Flask, TensorFlow, OpenCV, JUnit, Bootstrap, pandas, NumPy, Matplotlib

Developer Tools: Git, Docker, VS Code, Visual Studio, Microsoft Azure ML, Arduino IDE, ESP32, WordPress

Interests: Martial Arts, exploring new places, board games, 8-ball pool, soccer, F1, cycling, cinema